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GENERATING INTONATION FOR SWEDISH TEXT-TO-SPEECH CONVERSION USING A QUANTITATIVE MODEL FOR THE F_0 CONTOUR

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ABSTRACT

In Swedish, the fundamental frequency contour (F_0 contour) is known to be the main acoustic feature for word accent and sentence intonation. A model, based on an extension of the model for Japanese, is used for the generation process of F_0 contours of Swedish. As the input to this model, two kinds of command are assumed: the phrase commands which are positive impulses except at the end of an utterance, and accent commands which are stepwise functions of both polarity. Analysis-by-Synthesis of F_0 contours of both isolated words and sentences, uttered by two native speakers from the Stockholm region, indicated that the model can always generate very close approximations to observed F_0 contours, and that the extracted parameters are systematically related to the underlying lexical word accent, syntactic structure, and focus. Furthermore, the model is introduced into a framework of text-to-speech conversion for Swedish and an outline is given for the derivation of F_0 model parameters.

Keywords: *intonation, prosody, text-to-speech conversion, speech synthesis, fundamental frequency contour, analysis-by-synthesis.*

1. INTRODUCTION

In Swedish, as well as in other Scandinavian languages, the fundamental frequency contour (F_0 contour) is known to be the main acoustic feature for both word accent and sentence intonation. Although there already exist a number of studies on the analysis and modeling of F_0 contours, most of them are based either on qualitative observations [1] or on crude approximations to actually observed F_0 contours using piecewise linear functions [2], and therefore are not capable of producing accurate contours. A mathematically well-formulated model, which can generate the dynamic characteristics of the F_0 contour, is essential for realizing a high-quality text-to-speech synthesis system.

The idea of such a model has been shown by Öhman et al. [3], but has not been worked out in detail nor applied to speech synthesis. On the other hand, Fujisaki et al. [4] extended and elaborated Öhman's idea and presented a quantitative model, which was shown to be capable of closely approximating actually observed F_0 contours of Japanese words and sentences. It has been shown that Fujisaki's model can be successfully applied for both analysis and synthesis of F_0 contours of Japanese [5], Swedish [6] and several other languages.

In the present paper, we introduce Fujisaki's F_0-model into a framework of text-to-speech conversion for Swedish. The first stages of TTS conversion concerns the analysis of the input text and the subsequent determination of segmental and prosodic features. In the synthesis stage that information is used in the acoustic realization of the utterance including generation of the F_0 contour. A method for the derivation of F_0 model parameter values from analysis of the input text are outlined.

2. F_0-MODEL FORMULATION

Analysis of isolated words and sentences of Swedish suggests that the F_0 contour, when plotted on the logarithmic frequency scale as a function of time, can be regarded as the sum of a global component (henceforth phrase component) which initially rises and then gradually falls to approach zero, and one or more local accent components occurring at accented syllables whose shapes may either be plateau-like or peaked depending on the duration and whose polarity may either be positive or negative. Longer utterances may have additional phrase components starting a few hundred milliseconds earlier than the segmental onset of a major syntactic boundary. The accent components of several words may be merged into one when the words are not focused, but are quite prominent when the words are focused.

On the basis of these observations, we have proposed a model for F_0 contour generation for words and sentences of the Standard Swedish [6] as shown in Fig. 1. The phrase commands, here assumed to be impulses, generate the phrase components which constitute the global

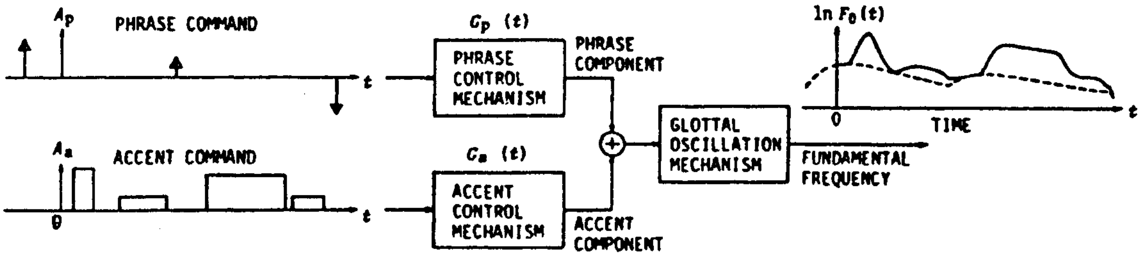

Figure 1. A functional model for generationg F_0 contours of sentences.

shape of the F_0 contour, and the accent commands, here assumed to be positive or negative stepwise functions, generate the local deviations from the global contour due to the presence of different word accents and sentence focuses. Both the phrase and accent control mechanisms are assumed to be critically damped second-order linear systems, and the sum of their outputs, i.e., the phrase components and accent components, is superposed on a baseline ($\ln Fb$) to form an F_0 contour. Thus an F_0 contour of a word or a sentence can be represented by the following equation:

$$\begin{aligned}\ln F_0(t) &= \ln Fb + \sum_{i=1}^{I} Ap_i Gp(t - T_{0i}) \\ &\quad + \sum_{j=1}^{J} Aa_j \{Ga(t - T_{1j}) - Ga(t - T_{2j})\},\end{aligned} \tag{1}$$

$$Gp(t) \begin{cases} = \alpha^2 t \exp(-\alpha t), & \text{for } t \geq 0, \\ = 0, & \text{for } t < 0, \end{cases} \tag{2}$$

$$Ga(t) \begin{cases} = \min[1 - (1 + \beta t)\exp(-\beta t), \gamma], \\ \qquad\qquad\qquad\qquad\qquad \text{for } t \geq 0, \\ = 0, \qquad\qquad\qquad\quad\;\; \text{for } t < 0, \end{cases} \tag{3}$$

where $Gp(t)$ represents the impulse response function of the phrase control mechanism and $Ga(t)$ represents the step response function of the accent control mechanism. The symbols in these equations indicate

- Fb : asymptotic value of fundamental frequency in the absence of accent components,
- I : number of phrase commands,
- J : number of accent commands,
- Ap_i : magnitude of the ith phrase command,
- Aa_j : amplitude of the jth accent command,
- T_{0i} : timing of the ith phrase command,
- T_{1j} : onset of the jth accent command,
- T_{2j} : end of the jth accent command,
- α : natural angular frequency of the phrase control mechanism,
- β : natural angular frequency of the accent control mechanism,
- γ : a parameter to indicate the ceiling level of the accent component (generally set equal to 0.9).

The parameters α and β are assumed to be constant, at least within an utterance, but are allowed to vary from utterance to utterance, and among speakers.

3. ANALYSIS-BY-SYNTHESIS OF F_0 CONTOURS

3.1. Procedure

The validity of the model can be tested by a procedure known as Analysis-by-Synthesis, i.e., by constructing the best approximation to an observed F_0 contour, and by examining the closeness of the approximation. In the present study, the optimization was carried out by minimizing the mean squared error in the $\ln F_0(t)$ domain through a hill-climbing search in the space of model parameters. This procedure allows one to decompose a given F_0 contour into its constituents, i.e. the phrase components and the accent components, and to estimate the magnitude and timing of their underlying commands by deconvolution.

3.2. Speech Material

In Swedish, polysyllabic words have two accent types (Accent 1 or Accent 2). The speech material for the present study consists of two sets. Set A contains three utterances each of the words *anden* (Accent 1, "the wild duck", henceforth to be denoted by *anden(1)*), and *anden* (Accent 2, "the spirit", henceforth *anden(2)*), both in isolation and in the carrier sentence *Nämna* ___ *igen* ("Mention ___ again"). Set B contains two utterances each of the sentence *Madame Marianne Malarmé har en mandolin från Madrid*, with varying focus positions. The informants were two male speakers from the Stockholm region. The recorded utterances were sampled at 10 kHz and digitized with 12 bit accuracy. The fundamental frequency was extracted by a modified autocorrelation method at every 10 msec.

3.3. Results of Analysis of Word Accent

Figures 2(a) and 2(b) respectively show one example each of the analysis results of the words *anden(1)* and *anden(2)* uttered by Speaker ML in the carrier sentence.

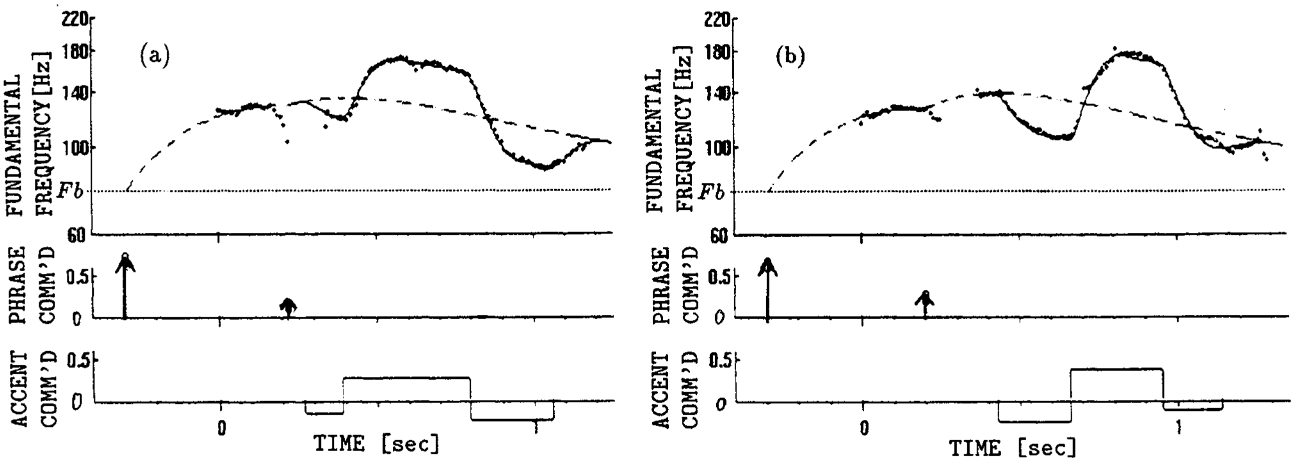

Figure 2. Analysis-by-Synthesis of F_0 contours of *anden* embedded in the carrier sentence *Nämna* ___ *igen.* The uppermost panel shows the measured F_0 contour, its closest approximation, and the phrase components. the middle and the lowermost panels indicate the phrase commands and the accent commands respectively. (a) *anden* (Accent 1), (b) *anden* (Accent 2).

	Fig 2(a)	Fig 2(b)	Fig 3
F_b (Hz)	78	78	78
A_p	0.7, 0.14	0.64, 0.22	0.6, 0.46
A_a	-0.16, 0.28, -0.24	-0.26, 0.38, -0.1	0.42, 0.26, -0.14,0.72,0.40
α (s^{-1})	2.0	2.0	1.7
β (s^{-1})	25	25	20
γ (s^{-1})	0.9	0.9	0.9

Table 1. Values of model parameters of best approximations shown in Figs. 2 and 3.

The curves shown by + symbols indicate the measured F_0 contour, the solid lines indicate the best approximations, and the broken lines indicates the estimated phrase components which are not directly observed in most cases. The difference between the solid line and the dotted line indicates the estimated accent components. The estimated phrase and accent commands are also shown in the lower part of each figure. These figures show that the model can generate very close approximations to observed F_0 contours. In both cases, a small second phrase command occurs at about 150 msec before the segmental onset of the target word (i.e. *anden(1)* or *anden(2)*). In both cases, the positive accent command is extended to the unstressed initial syllable of the following word *igen* (accent 1), followed by a negative accent command occurring at the beginning of the stressed second syllable of *igen.* The same type of negative accent command is sometimes present as a small negative command preceding the positive one, as is the case for *anden(1).* This pattern is consistent with the rest of the speech material studied and agrees well with previously reported work [7]. The values of the model parameters for these examples are given in Table 1 together with the parameters for the sentence shown in Fig. 3.

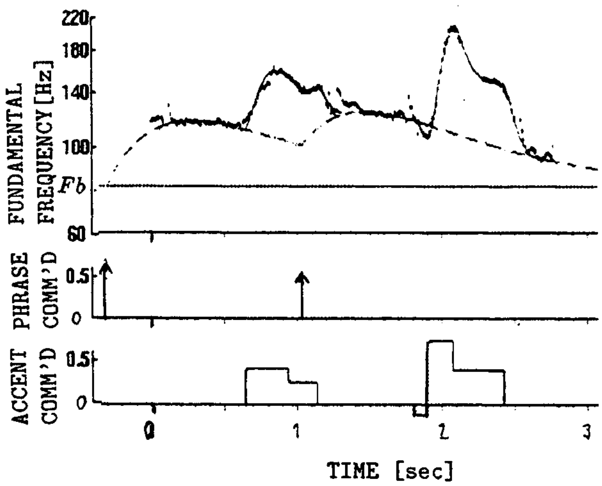

Figure 3. Analysis-by-Synthesis of the F_0 contour of the sentence *Madame Marianne Malarmé har en mandolin från Madrid* with a focus on *"mandolin".*

3.4. Results of Analysis of Sentence Intonation

Figure 3 shows an example of the results of analysis of a sentence from Set B, again by Speaker ML. The intended focus is on the word *mandolin.* The results indicate the existence of a large second phrase command occurring at a few hundred milliseconds before the segmental onset of the predicate phrase (i.e., *har en mandolin från Madrid*). The intended focus is manifested by a major accent command for the third syllable of *mandolin,* which is followed by a somewhat smaller accent command continuing until the accented second syllable of the word *Madrid.* There is also a fairly large accent command for the second syllable of *Marianne,* continuing to a somewhat smaller accent command until the accented third syllable of *Malarmé.*

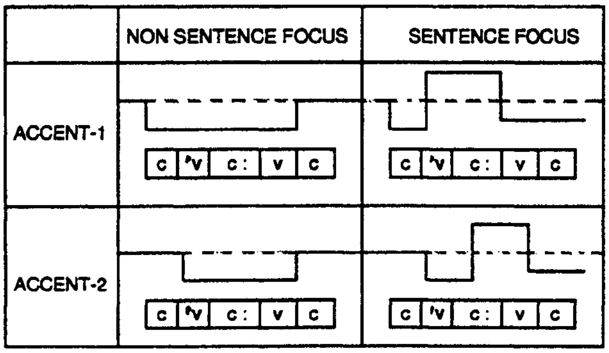

Figure 4. Accent command patterns for Accent 1 and Accent 2 words in focus and non-focus position (c - consonant, v - vowel, 'v - vowel with primary stress).

4. GENERATING INTONATION FOR TEXT-TO-SPEECH CONVERSION

The first stage in the text-to-speech conversion process involves various types of linguistic processing, such as text-normalization, lexical look-up, syntactic analysis and semantic/pragmatic analysis. This processing provides the segmental and prosodic feature specifications necessary for generating natural sounding synthetic speech. The prosodic specification includes:

- lexical accent of constituent words,
- marking of stressed syllables,
- marking of sentence focus (degree of prominence)
- sentence and phrase boundaries.

Having access to prosodic features of this type and knowledge on how they are realized acoustically in the language, one may now generate synthesizer specifications for the F_0 contour, segment durations, and amplitudes. In this discussion we focus on the F_0 contour generation.

The analysis results of F_0 contours of Swedish utterances, shown in Figs. 2 and 3, correspond well with the rest of the collected speech material as well as with previously reported work [8]. Based on these results we conjecture prototypical patterns for accent command timing, valid for words carrying accent 1 and accent 2, with and without sentence focus, as shown in Fig. 4 for CVCVC words.

As was noted above, the focal accent command often extends into the following words. For our purposes, we design the following rule: *The "return phase" of the accent command for words with sentence focus, occurs in conjunction with the accent command of the next stressed syllable.*

The amplitudes of the accent commands are related to the degree of stress (prominence) associated with the constituent words.

Phrase commands are inserted three hundred milliseconds earlier than the segmental onset of major syntactic boundaries

Within the same utterance type and speaker, the parameters F_b, α, β and γ could be set to constant values.

5. DISCUSSION AND CONCLUSION

The closeness of approximation, obtained by Analysis-by-Synthesis of F_0 contours of these and many other utterances of Swedish, attests the basic validity of the proposed model for the process of generating F_0 contours of Swedish. It is not surprising that a model for the F_0 contours of Japanese can in principle be extended to F_0 contours of Swedish, since the model is considered to represent the dynamic characteristics of the control mechanism of the voice fundamental frequency which is more or less language-independent. Moreover, the language-specific systematic relationships found between the input parameters, such as the timing and the magnitude of the phrase and accent commands, and the underlying lexical word accent, syntactic structure and focus indicate the model's utility in speech synthesis of Swedish. From the viewpoint of speech synthesis by rule, an attempt was made at formulating basic rules based on these relationships.

Further investigation is obviously necessary to test the validity of the model for a wider variety of utterances, of speakers, and of dialects.

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